Digital Parking Management

Scenario

- The city operates multiple parking zones with charging points
- Citizens can search, reserve, and use parking spaces
- The city administration manages:
  - Parking zones
  - Parking spaces
  - Pricing policies
- Your team will design and implement the cloud backend as a modular monolith

Scenario

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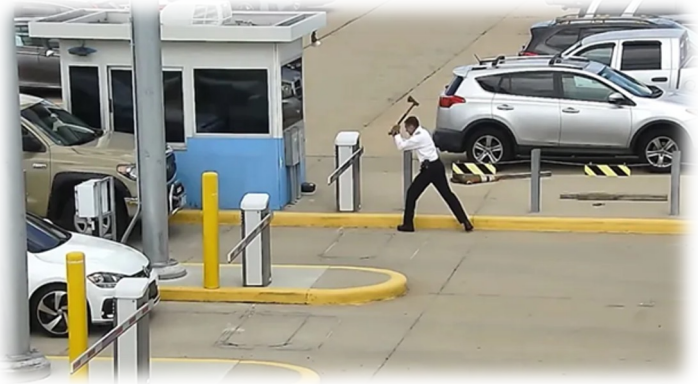
- The system enables:
  - Citizens to find and reserve parking spaces
  - The city to manage zones and monitor occupancy
  - Billing for parking usage and for eletric charging



Objective

Design and implement a Modulith-based backend that:

- Supports parking zone management
- Handles reservations
- Ensures consistency of parking space states

- You must:
  - Derive a clean architecture using the C4 model
  - Define clear module boundaries
  - Implement core functionality and provide an UI
  - Avoid tight coupling between modules

Functional Requirements

- Parking zone management (create, update, delete)
- Pricing and billing (parking space and charging)
- Parking space management (free, reserved, occupied)
- Parking space search & reservation
  - Includes also visualization via a map
- User management
- Feel free to add additional functionality that improves the user experience or ease the management, e.g.,
  - the system automatically recognizes the vehicle via the number plate
  - the system calculates the cheapest charging strategy

Non-functional Requirements

- Modultih architecture with single deployable backend
- Clear module boundaries
- No direct database access across modules
- Shared database with representative data
- Communication via Interfaces and Domain events
- User interface for both citizens and city administrators

- Optional: Use Spring Moduliths features such as module integration testing, diagram generation, etc.

Expected Outcomes

- Each team must provide:
  - C4 Context Diagram
  - C4 Container Diagram
  - C4 Component Diagram (modules)
  - Short explanation of module responsibilities
  - Working prototype
- For the final presentation also show your project management infrastructure and how you worked as a group